

Samarth Gowda

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PROFESSIONAL OBJECTIVE

Electrical and Electronics Engineer passionate about hardware design, embedded systems, VLSI, and low-level software. Seeking to bridge physical silicon and creative engineering.

EDUCATION

International Institute of Information Technology Bangalore (IIITB)

03/2026 – 07/2026

Post Graduate Diploma in VLSI Design & Verification | Bengaluru, India

Government Engineering College Mosalehosahalli

12/2022 – 06/2026

B.E. in Electrical and Electronics Engineering | CGPA: 7.9/10 | Hassan, Karnataka
President & Founder of ELECTROSPHERE & VEDA (VLSI & Embedded Design Association)

TECHNICAL SKILLS

Hardware & EDA: Verilog HDL, SystemVerilog, Digital Circuit Design, KiCAD, FreeCAD, Xilinx Vivado, MATLAB

Embedded & Systems: C, C++, ESP32, STM32, Arduino, Linux (Fedora KDE/Pop!_OS), Bash, FreeRTOS

Creative & Software: Python, DaVinci Resolve (Video Editing), Git, Web Technologies, Blender

EXPERIENCE

Nexus Core Innovations — Design & Verification Intern

03/2026 – 06/2026

- Conducted VLSI design verification and simulation of digital modules in Verilog.
- Wrote automated testbenches and validated functional corner cases.

Multimedia Creator & Video Editor — Independent / Remote

03/2025 – Present

- Produced high-retention video content using DaVinci Resolve, focusing on storytelling and visual pacing.
- Managed end-to-end post-production, asset curation, color grading, and audio mixing.

SELECTED ENGINEERING PROJECTS

Solar-Powered EV Charging Station with Battery Management System (BMS)

- Integrated Li-ion storage, micro-controller, and solar PV array for sustainable EV charging.
- Implemented real-time telemetry algorithms for battery health, temperature, and charge cycles.

bibliothek — Modern C++ Library Management System

- Engineered robust CLI application with persistent storage, clean OOP design, and algorithms.

Humble Beginnings: C Systems Programming & Low-Level Labs

- Implemented custom memory allocators, pointer arithmetic routines, and core data structures.

CERTIFICATIONS & HONOURS

- Advanced Drone Technology (Air Taxi) Workshop — Bharat Space Education Research Centre
- Summer School on Digital Hardware Design with Verilog — IEEE